

Economic Growth II

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The Solow Model: Golden Rule

- We studied the conditions for a steady state equilibrium
- Is there a way to rank steady state equilibria with some welfare criterion?
- By introducing a normative dimension we can thus answer the question: how much ***should*** the economy save?
- **One** possible criteria is the **Golden Rule**: Max. steady state consumption

$$C_{ss} = Y_{ss} - S_{ss} = (1 - s) \cdot Y_{ss}$$

$$c_{ss} = (1 - s) \cdot y_{ss}$$

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The Solow Model: Golden Rule

- $Max. c_{ss}$ comes by balancing out two effects:
 - ▶ Higher s has direct effect of lowering c_{ss}
 - ▶ But higher s increases k_{ss} and thus y_{ss} and c_{ss}
- Let's restate the problem as $Max. c_{ss}$ by choosing k_{ss} (instead of s)
- We now have to make s a function of k_{ss} to then use this to $Max. c_{ss}$
- In steady state we know that:

$$sf(k_{ss}) = (\delta + n)k_{ss}$$

Thus:

$$s = \frac{(\delta + n)k_{ss}}{f(k_{ss})}$$

which we can use in the equation for c_{ss} .

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The Solow Model: Golden Rule

$$\begin{aligned} c_{ss} &= \left(1 - \frac{(\delta + n)k_{ss}}{f(k_{ss})}\right) f(k_{ss}) \\ &= f(k_{ss}) - (\delta + n)k_{ss} \end{aligned}$$

- $Max. c_{ss}$: first order condition:

$$f'(k_{gr}) = \delta + n$$

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The Solow Model: Golden Rule

- For the **Cobb-Douglas** case:

$$\alpha k_{gr}^{\alpha-1} = \delta + n$$
$$k_{gr} = \left(\frac{\delta + n}{\alpha} \right)^{\frac{1}{\alpha-1}}$$

- Replacing this into the formula for s , we get:

$$s = \frac{(\delta + n) \left(\frac{\delta + n}{\alpha} \right)^{\frac{1}{\alpha-1}}}{\left(\frac{\delta + n}{\alpha} \right)^{\frac{\alpha}{\alpha-1}}}$$
$$= \alpha$$

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The Solow Model: Factor Markets

- Behind the mechanics of the model: firms and workers interacting in markets
- Factor Markets are assumed to be **perfectly competitive**
- **Labor market:** workers are paid a real wage w per unit of labor supplied; firms hire as many units of work they want at wage w ; in equilibrium w is such that firms want to hire exactly L
- **Capital market:** firms rent capital from households (firms' owners) for one period, paying r^K and returning depreciated capital to households; in equilibrium r^K is such that firms rent exactly K that is available

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The Solow Model: Factor Markets

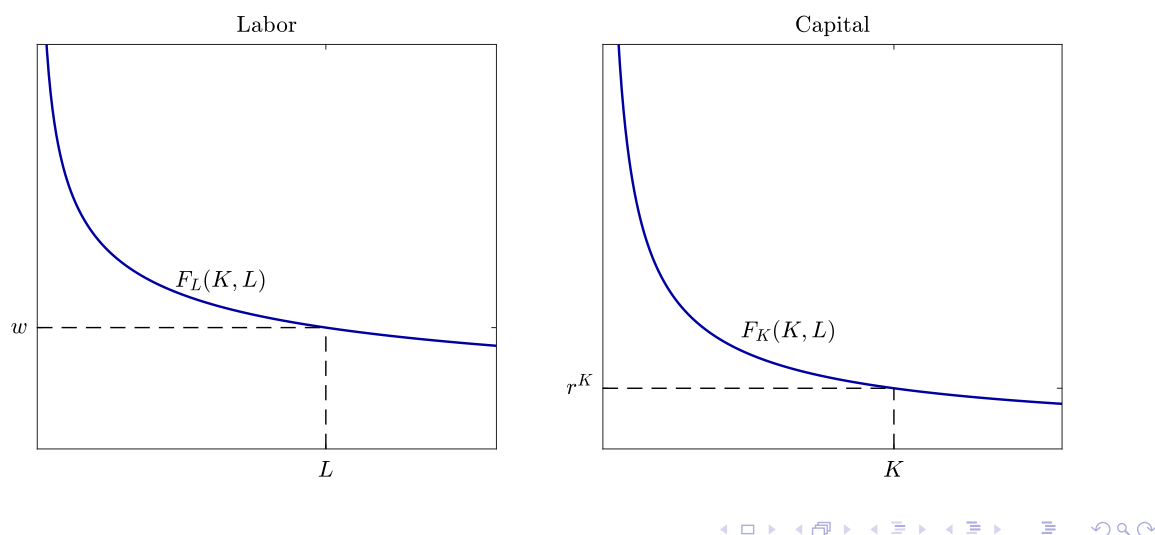
- **Firm's Decision:**

$$\max_{K_i, L_i} F(K_i, L_i) - wL_i - r^K K_i$$

- **First Order Conditions:**

$$F_K(K_i, L_i) - r^K = 0 \quad (1)$$

$$F_L(K_i, L_i) - w = 0 \quad (2)$$



The Solow Model: Factor Markets

- **Market Clearing:** we rely on the following mathematical results

Theorem (1)

Euler's Theorem

$$F_K(K, L)K + F_L(K, L)L = F(K, L) \quad (3)$$

Proof.

From constant returns to scale:

$$F(\lambda K, \lambda L) = \lambda F(K, L)$$

Derive with respect to λ :

$$F_K(\lambda K, \lambda L)K + F_L(\lambda K, \lambda L)L = F(K, L)$$

Set $\lambda = 1$.



The Solow Model: Factor Markets

- **Market Clearing:** we rely on the following mathematical results

Proposition (1)

Production Function Satisfies

$$F_K(\lambda K, \lambda L) = F_K(K, L) > 0 \quad \forall \lambda > 0 \quad (4)$$

$$F_L(\lambda K, \lambda L) = F_L(K, L) > 0 \quad \forall \lambda > 0 \quad (5)$$

Proof.

From constant returns to scale:

$$F(\lambda K, \lambda L) = \lambda F(K, L)$$

Derive with respect to K :

$$F_K(\lambda K, \lambda L)\lambda = \lambda F_K(K, L)$$

Do the same with respect to L .



The Solow Model: Factor Markets

These mathematical results imply that all firms in the economy use factors in the same proportions

- Take Prop. (1) for a given firm i and set $\lambda = \frac{1}{L_i}$ in (4) and $\lambda = \frac{1}{K_i}$ in (5):

$$F_K(K_i, L_i) = F_K\left(\frac{K_i}{L_i}, 1\right) \quad (6)$$

$$F_L(K_i, L_i) = F_L\left(1, \frac{L_i}{K_i}\right) \quad (7)$$

- So marginal products depend on the ratio of production factors, and not on their absolute values



The Solow Model: Factor Markets

First order conditions (1) and (2) plus perfect competition imply all firms face the same factor prices and thus have the same marginal product

- Therefore: all firms choose the same ratio of capital to labor
- Without loss of generality, assume there is one **representative firm** doing all production, implying that:

$$r^K = F_K(K, L) \quad (8)$$

$$w = F_L(K, L) \quad (9)$$

- From (6) and (7) the above equations become:

$$r^K = F_K(k, 1) \quad (10)$$

$$w = F_L\left(1, \frac{1}{k}\right) \quad (11)$$

- An economy with high k has low rental rate and high wages

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The Solow Model: Factor Markets

Profits (real terms) of the representative firm: $\Pi = F(K, L) - wL - r^K K$

Proposition (2)

The representative firm earns $\Pi = 0$.

Proof.

Just use (8) and (9) to replace marginal products in the Euler equation (3). ◻

- But “profits” here mean **economic profits** instead of accounting profits

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The Solow Model: Factor Markets

Example

- Plantain Monarchy is a corporation that owns a retail space where it runs a clothes shop
- In 2018 it sold \$1,000,000 in clothes, paid \$300,000 in wages and bought clothes worth \$500,000
- Rent on comparable retail space is \$ 200,000 a year
- What were the corporation's profits?

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The Solow Model: Factor Markets

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Income Statement (Accounting)		
	Sales	1, 000, 000
-	Cost of Goods Sold	500, 000
-	Expenses	300, 000
=	Profits	200, 000

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The Solow Model: Loan Market and Interest Rate

- Return on investing $\times$ goods:

$$\begin{array}{rcl} & \text{Rental Income} & r_{t+1}^K x \\ + & \text{Value of depreciated capital} & (1 - \delta) x \\ \hline = & \text{Total} & (1 + r_{t+1}^K - \delta) x \end{array}$$

- Return on lending $\times$ goods:

$$(1 + r_{t+1})x$$

- Indifference requires this condition to hold:

$$r_{t+1} = r_{t+1}^K - \delta = F_K(k_{t+1}, 1) - \delta \quad (12)$$

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The Solow Model: Prices

- Equations (10), (11) and (12) tell us what happens when steady state changes
- **Example 1:** we've studied the case of an exogenous increase in the saving ratio, s , and know it leads to a higher level of k_{ss} . Thus it also implies that wages will be higher and r^K and r will be lower.
- **Example 2:** we've studied the case of an exogenous increase in the population growth rate, n , and know it leads to a lower level of k_{ss} . Thus it also implies that wages will be lower and r^K and r will be higher.

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The Solow Model: Technological Progress

- As we did before, we now need to get the dynamics of capital per efficiency unit of labor (instead of per capita), $\Delta \tilde{k}_{t+1}$

$$\begin{aligned}
 \Delta \tilde{k}_{t+1} &\equiv \tilde{k}_{t+1} - \tilde{k}_t = \frac{K_{t+1}}{A_{t+1}L_{t+1}} - \tilde{k}_t \\
 &= \frac{(1-\delta)K_t + I_t}{A_{t+1}L_{t+1}} - \tilde{k}_t = \frac{(1-\delta)K_t + sY_t}{A_{t+1}L_{t+1}} - \tilde{k}_t \\
 &= \left[\frac{(1-\delta)K_t + sY_t}{A_t L_t} \right] \frac{A_t L_t}{A_{t+1} L_{t+1}} - \tilde{k}_t \\
 &= [(1-\delta)\tilde{k}_t + s\tilde{y}_t] \frac{1}{(1+n)(1+g)} - \tilde{k}_t \\
 &= \frac{s\tilde{y}_t - (\delta + n + g + ng)\tilde{k}_t}{1 + n + g + ng} \\
 &\approx \frac{s\tilde{y}_t - (\delta + n + g)\tilde{k}_t}{1 + n + g} \\
 &= \frac{sf(\tilde{k}_t) - (\delta + n + g)\tilde{k}_t}{1 + n + g}
 \end{aligned} \tag{13}$$

The Solow Model: Technological Progress

- If variables per efficiency unit of labor are constant in steady state, does this mean technological progress is irrelevant?
- No!**
- Economic growth is about per capita variables!
- $\tilde{y}$ constant in steady state implies:

$$\begin{aligned}
 \tilde{y}_{t+1} &= \tilde{y}_t \\
 \frac{y_{t+1}}{A_{t+1}} &= \frac{y_t}{A_t} \\
 \frac{y_{t+1}}{y_t} &= \frac{A_{t+1}}{A_t} = (1+g)
 \end{aligned}$$

The Solow Model: Technological Progress

Proposition

- ① In steady state the following variables grow at the rate of technological progress:
 - ▶ GDP per capita, y
 - ▶ Capital per worker, k
- ② In steady state, the ratio capital-output (K/Y) is constant

Proof.

- ① Already proved for y and we prove for k in similar way
- ② Follows from item (1) of this proposition:

$$\frac{K_{t+1}}{Y_{t+1}} = \frac{k_{t+1}}{y_{t+1}} = \frac{k_t(1+g)}{y_t(1+g)} = \frac{K_t}{Y_t}$$



Navigation icons: back, forward, search, etc.

Confronting Theory and Evidence: The Solow Model

- How does the Solow Model with Technological Progress fare with respect to the “Kaldor Facts”?
 - ① “GDP per capita grows at constant rate” ✓ (by assumption)
 - ② “The ratio of total capital stock to GDP is constant” ✓
 - ③ “The shares of labor and capital income in GDP are constant”

$$\begin{aligned}\frac{wL}{Y} &= \frac{\frac{\partial F(K, AL)}{\partial L} L}{F(K, AL)} = \frac{AF_L(K, AL)L}{F(K, AL)} \\ &= \frac{AF_L\left(\frac{K}{AL}, 1\right) L}{F\left(\frac{K}{AL}, 1\right) AL} = \frac{F_L\left(\frac{K}{AL}, 1\right)}{F\left(\frac{K}{AL}, 1\right)}\end{aligned}$$

Euler’s Theorem tells us that labor and capital income shares add up to 1, so capital share is also constant ✓

Exercise: show that for the Cobb-Douglas production function income shares are also constant off steady state

Navigation icons: back, forward, search, etc.

Confronting Theory and Evidence: Putting Numbers on the Solow Model

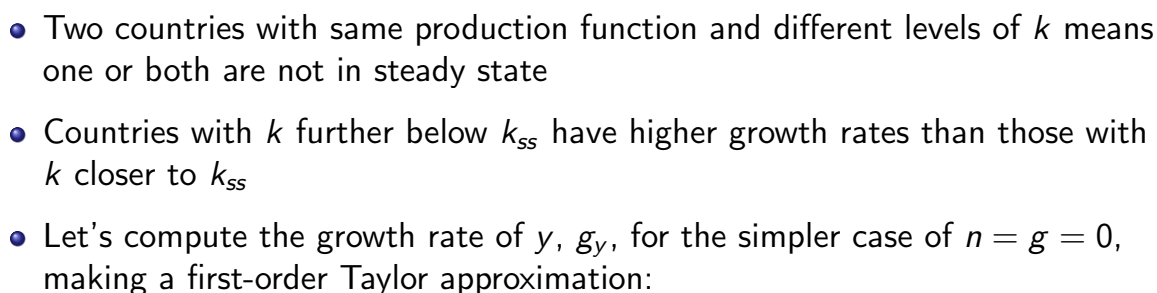
- With these parameters we get $r = F_K(K, AL) - \delta = \alpha \frac{K}{AL} - \delta = \alpha \frac{\delta+n+g}{s} - \delta \approx 0.074$ which is higher than average rate on 10-year inflation-indexed bonds (1 or 2%); model has no risk
- We can also simulate the model for an once-and-for-all increase in the parameters: see the book (pp. 79-80) for an increase in s



Confronting Theory and Evidence: The Capital Accumulation Hypothesis

- If technology levels are the same across countries, (Smith's) question about why some countries are rich while others aren't would have as answer differences in K/L
 - ▶ Then poor countries would need to foster capital accumulation
- Evidence rejects this conjecture, and we can address this looking to different things
 - ① Convergence
 - ② Direct Measurement
 - ③ Interest Rate and Capital Flows [will not discuss in class, see the book]

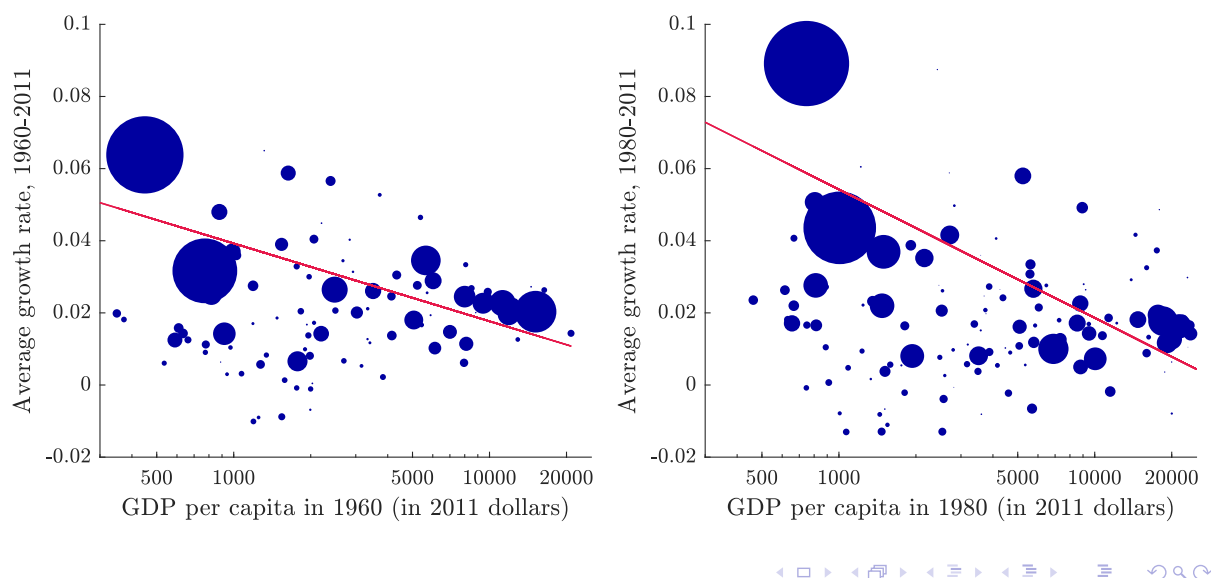




Confronting Theory and Evidence: The Capital Accumulation Hypothesis

(1) CONVERGENCE:

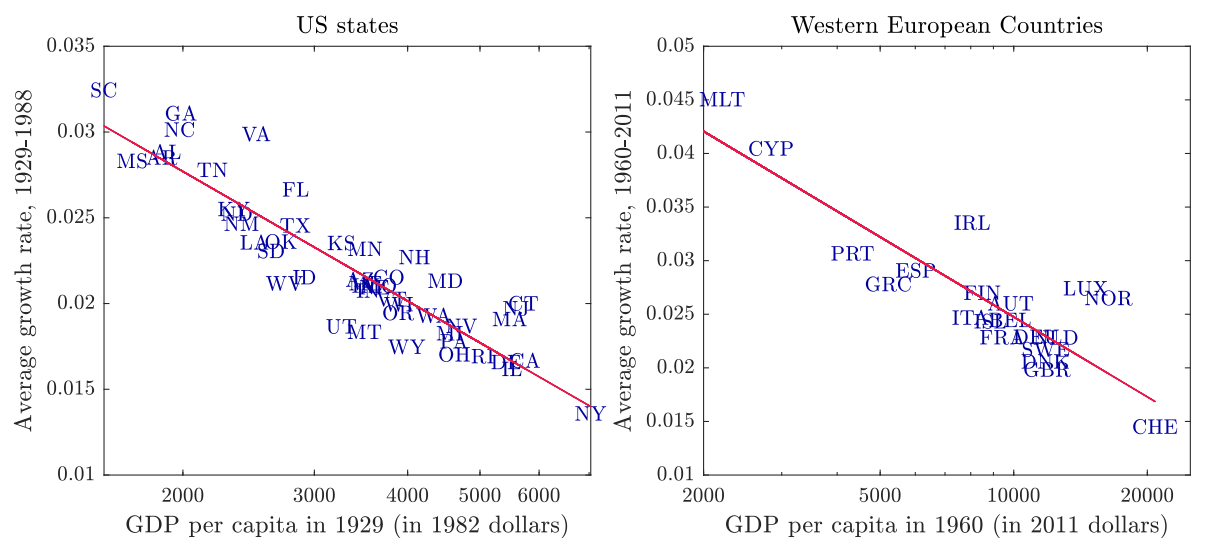
- But do all countries count the same in such correlation? What about weighting them by population?



Confronting Theory and Evidence: The Capital Accumulation Hypothesis

(1) CONVERGENCE:

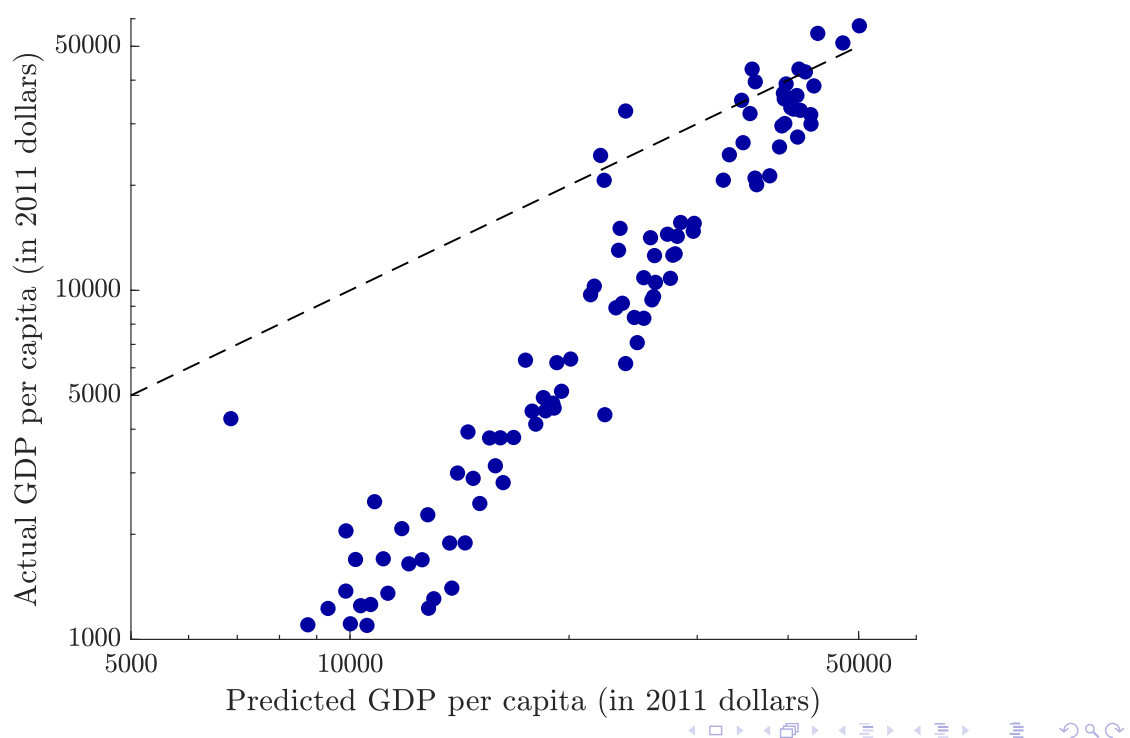
- What about specific groups of economies or states of a country?



Confronting Theory and Evidence: The Capital Accumulation Hypothesis

(2) DIRECT MEASUREMENT:

- If we:
 - ➊ Measure k directly in each country
 - ➋ Assume a functional form for the production function, e.g. $f(k) = Ak^\alpha$, setting $\alpha = 0.35$
 - ➌ Solve from data for $A = \frac{y_{USA}}{k_{USA}^\alpha}$ to then predict y for each country: $\hat{y}_j = Ak_j^\alpha$
 - ➍ Compare predicted and measured GDP (that would have to look similar under our hypothesis being tested)



Confronting Theory and Evidence: Growth Accounting

- Growth Accounting is a technique that decomposes GDP growth into the growth of K , L , and A (all possible sources)
- Basic idea:
 - 1 Measure how much K and L have changed
 - 2 Infer how much Y should have changed based on that (so we are combining theory and measurement)
 - 3 Attribute all changes in measured Y that cannot be attributed to K and L to changes in A

$$Y_{t+1} = F(K_{t+1}, L_{t+1}, A_{t+1}) \approx F(K_t, L_t, A_t) + F_K(K_t, L_t, A_t)(K_{t+1} - K_t) + F_L(K_t, L_t, A_t)(L_{t+1} - L_t) + F_A(K_t, L_t, A_t)(A_{t+1} - A_t)$$

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Confronting Theory and Evidence: Growth Accounting

- We thus obtain the growth rate:

$$g_Y \equiv \frac{Y_{t+1}}{Y_t} - 1 = \frac{F_K(K_t, L_t, A_t)K_t}{F(K_t, L_t, A_t)} \cdot \frac{(K_{t+1} - K_t)}{K_t} + \frac{F_L(K_t, L_t, A_t)L_t}{F(K_t, L_t, A_t)} \cdot \frac{(L_{t+1} - L_t)}{L_t} + \frac{F_A(K_t, L_t, A_t)A_t}{F(K_t, L_t, A_t)} \cdot \frac{(A_{t+1} - A_t)}{A_t}$$

- The last term is the so-called “**Solow Residual**” (the unknown; also known as **Total Factor Productivity**) and the terms pre-multiplying g_K and g_L are capital and labor shares

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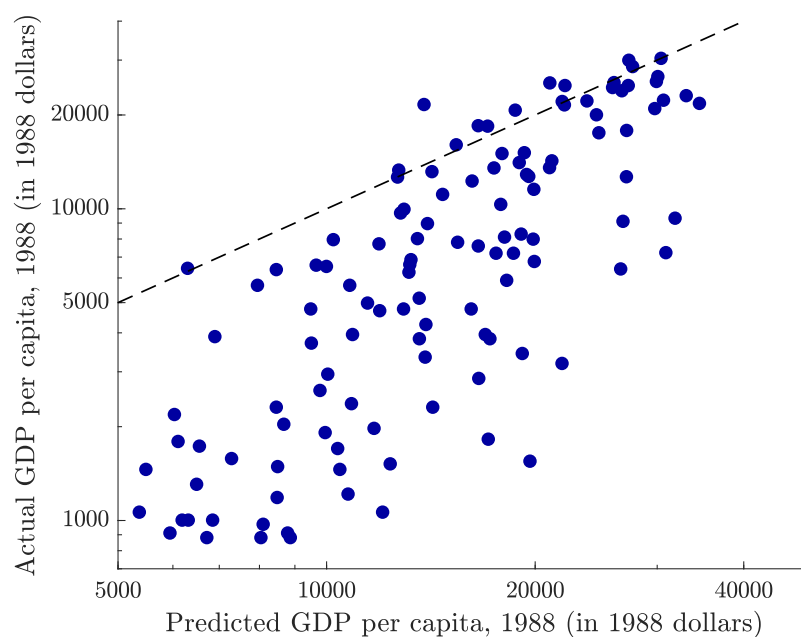
Confronting Theory and Evidence: Sources of TFP Differences

- What explains TFP differences across countries?
 - ④ **Human Capital:** workers are not homogeneous, but differ in knowledge, skills or talent; it can be accumulated through education
 - ▶ Hall and Jones (1999): directly measured K and computed a human-capital-adjusted measure of L : $\text{Wage} = w \cdot (\text{Years of Education})$
 - ▶ Human capital helps explain cross-country differences in productivity, but large differences remain

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Confronting Theory and Evidence: Sources of TFP Differences

1 Human Capital:



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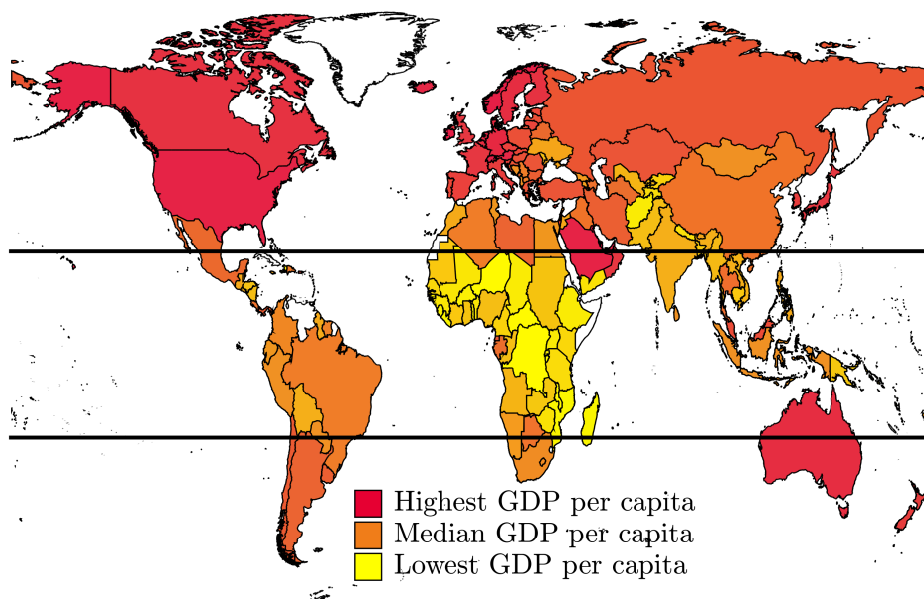
Confronting Theory and Evidence: Sources of TFP Differences

- What explains TFP differences across countries?
 - ② **Geography:** countries in colder climates tend to be richer than those in hotter climates
 - ▶ Sachs (2001): climate in itself matters –crop yields, diseases



Confronting Theory and Evidence: Sources of TFP Differences

2 Geography:



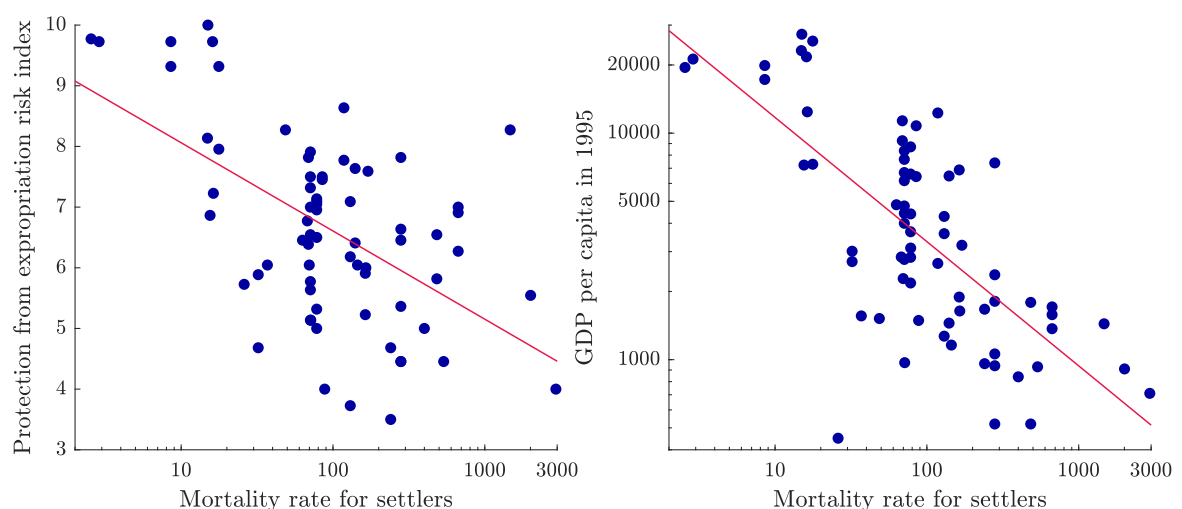
Confronting Theory and Evidence: Sources of TFP Differences

- What explains TFP differences across countries?
 - ③ **Institutions:** a host of political and social conditions that vary across countries (democracy, rule of law, political transparency, respect for property rights, etc.)
 - ▶ Institutions have large effects on productivity
 - ▶ Causality problem
 - ▶ Acemoglu, Johnson and Robinson (2001): Institutions do cause TFP; focused on countries with European colonization
 - ▶ They look at the correlation between mortality rate of European settlers and present-day institutions (risk of expropriation)
 - ▶ Low-mortality places where places they wanted to settle and live, and thus brought more open institutions developed in Europe
 - ▶ High-mortality places came with more “extractive” institutions, using political and military powers to extract resources

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Confronting Theory and Evidence: Sources of TFP Differences

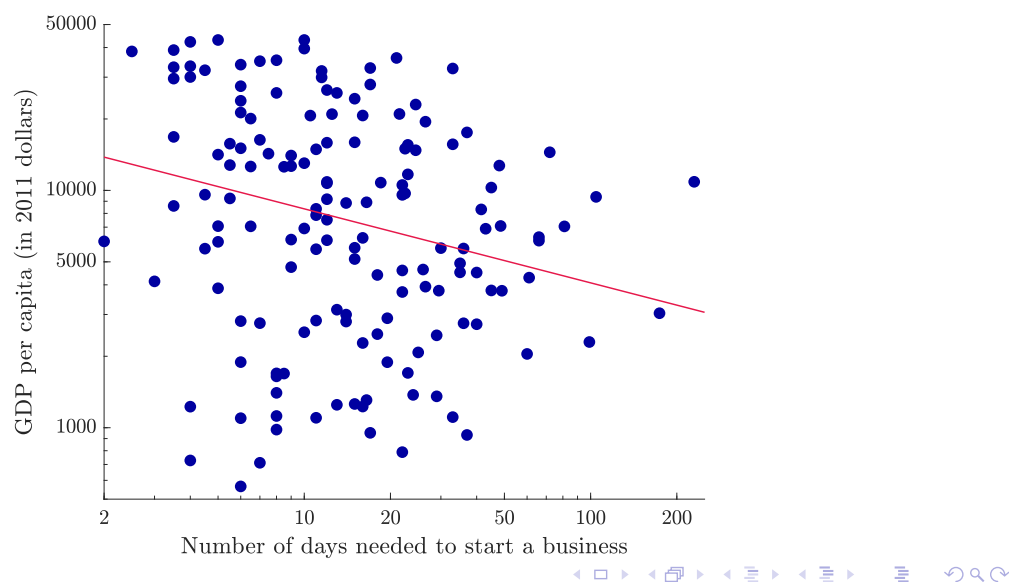
3 Institutions:



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Confronting Theory and Evidence: Sources of TFP Differences

- What explains TFP differences across countries?
 - ④ **Resource Allocation and Misallocation:** things that prevent K and L be put to good use
 - ▶ Entry regulation (starting a business)



Confronting Theory and Evidence: Sources of TFP Differences

- What explains TFP differences across countries?
 - ④ **Resource Allocation and Misallocation:** things that prevent K and L be put to good use
 - ▶ Entry regulation (starting a business)
 - ▶ Lack of credit
 - ▶ Restrictions on Foreign Investment
 - ▶ Monopoly rights
 - ▶ Uneven taxation or regulation across firms